

Fault Code: 442

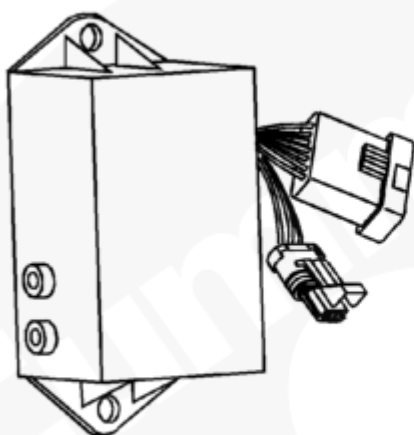
Battery Voltage Out of Range - High

 Printable Version

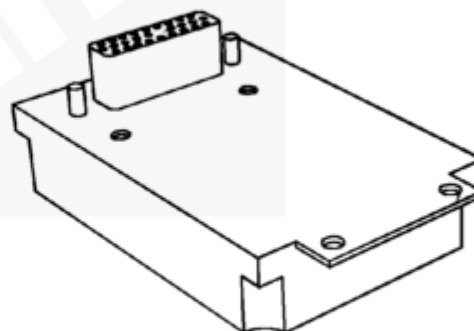
Overview

Codes	Reason	Effect
Fault Code: 442 PID(P): SPN: FMI: Lamp: Red SRT:	Battery Voltage Out of Range - High. Battery voltage is more than the high threshold detected by the Centinel™ control module.	The Centinel™ control module voltage supply approaching a level at which unpredictable operation will occur.

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HD



HHP

05800058

Circuit Description

The Centinel™ control module receives unswitched battery voltage from the starter for heavy-duty. For high-horsepower, the Centinel™ control module receives switched power from the fuel shutoff valve solenoid. There are two in-line 5-amp fuses in the Centinel™ control module heavy-duty harness to protect the Centinel™ control module. The high-horsepower has **only** one 5-amp fuse. The battery return wires in the engine harness are connected to the engine block ground.

Component Location

The location of the battery will vary with the OEM. Refer to the OEM manual for the battery location.

Shoptalk

This fault can be caused by loose or corroded battery connections.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the equipment battery system.	
	STEP 1A. Inspect the battery cable connections.	No damaged connections
	STEP 1B. Check the battery voltage.	Heavy-duty: 9 to 32 VDC 12-VDC High-horsepower: 8.2 - 17.3 VDC 24-VDC High-horsepower: 15.5 - 30.3 VDC
	STEP 1C. Check the Centinel™ control module battery voltage.	
STEP 2.	Clear the fault code.	
	STEP 2A. Disable the fault code.	Fault Code 442 inactive

STEP 1. Check the equipment battery system.

STEP 1A. Inspect the battery cable connections.

Conditions :

- Turn the keyswitch OFF.

Action	Specification/Repair	Next Step
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- Corrosion - Loose connection.	No damaged connections	1B
	Repair or replace the damaged connections - Repair the battery or starter connections. - Replace the battery or starter connections. Refer to the OEM service manual.	2A

STEP 1B. Check the battery voltage.

Conditions : Turn the keyswitch ON.		
Action	Specification/Repair	Next Step
Measure the battery voltage.	Heavy-duty: 9 - 32 VDC 12VDC High-horsepower: 8.2 - 17.3 VDC 24VDC High-horsepower: 15.5 - 30.3 VDC	1C
	Replace the battery. Refer to the OEM service manual.	2A

STEP 1C. Check the Centinel™ control module battery voltage.

Conditions : Turn the keyswitch ON.		
Action	Specification/Repair	Next Step

- Heavy-duty: Measure the voltage between pins 1 (+) and 2 (-) of the Centinel™ control module harness connector. - High-horsepower: Measure the voltage between pins 22 (+) and 25 (-) of the Centinel™ control module connector.	Reading must match Step 1B.	Complete
	Replace the Centinel™ wiring harness. Refer to Procedure 019-131 (/qs3/pubsys2/xml/en/procedures/96/96-019-131-tr.html).	2A

STEP 2. Clear the fault code.

STEP 2A. Disable the fault code.

Conditions :

- Connect all the components.

Action	Specification/Repair	Next Step
- Connect all the components. - Start the engine and let it idle for 1 minute. - Verify that Fault Code 442 is inactive.	Fault Code 442 inactive	Complete
	Return to the troubleshooting step or contact your local Cummins Authorized Repair Location if all the steps have been completed and checked again.	1A